Few biological parameters suffice to reproduce measured dynamics: verified parameter subset selection for systems biology models

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Abstract

Models of biological processes contain many unknown rate constants, and data rarely suffice to estimate them all. We present a parameter subset selection method that, starting from published values, finds a small set of biological parameters sufficient to reproduce the measured outputs when only they and the measurement scaling and offset factors are re-estimated. Parameters are ranked by their effect on the outputs, redundant ones are rejected, the reduced model is tested by re-estimation under random perturbations and unneeded parameters are removed. In 31 of 34 published models with real data a subset, usually of two biological parameters, was sufficient, and most remained adequate for parameter changes of up to 50%. With the other parameters at their published values, all selected parameters were practically identifiable in 25 of the 31 models. Among five criteria selecting equally many parameters, only ours reproduced the full model in all 28 comparable models (14 to 26 for the others), and in a leakage-free synthetic test our subsets reached the prediction threshold more often than those of pivoted QR, profile likelihood and Morris screening. The mathematical steps, including a sufficient condition for identifiability, are verified in the Lean proof assistant.

Keywords: parameter reduction, practical identifiability, sensitivity analysis, ordinary differential equations, systems biology, formal verification

Introduction

Ordinary differential equation (ODE) models are central to quantitative biology, pharmacology and epidemiology, where they describe how interacting molecular species or populations change over time as a function of kinetic parameters θ[1]. Using a model for prediction requires estimating θ from experimental measurements, and this becomes difficult as models grow. The main obstacle is non-identifiability: different parameter values may produce the same measured output, either because of model symmetries (structural non-identifiability) or because the available data are insufficient (practical non-identifiability)[2–4]. In both cases the likelihood has flat ridges on which optimisers stall and confidence intervals become unreliable[5–7]. Models in public benchmark collections routinely contain 20–300 parameters[8], and genome-scale models contain thousands[9].

The sloppy-model phenomenon offers a structural explanation[10]: the sensitivity of model outputs to parameters is typically dominated by a few "stiff" directions, while many "sloppy" directions have almost no effect on the outputs. This suggests that a small subset of parameters may suffice to describe what is actually measured. Exploiting this observation in practice requires answering three questions that existing tools address only partially. First, which parameters should be retained? Many criteria exist: global sensitivity indices[11, 12] and profile likelihoods[5, 29] rank parameters by importance or identifiability, and subset-selection methods based on the sensitivity matrix—pivoted QR[32], collinearity indices[14] and related identifiability screenings[30]—additionally avoid redundant parameters[13]. All of them return a subset; profile-likelihood-based reduction[29] assesses the reduced model through the likelihood of the data, but, to our knowledge, these criteria do not test whether re-estimating the selected subset reproduces the outputs of the full model when all parameters are perturbed. Second, is the reduced model good enough? A reduction is useful only if re-estimating the retained parameters reproduces the measured data and predicts experiments that were not used for fitting. Third, how much can the conclusions be trusted? Statements such as "the discarded parameters can be fixed" rest on approximations whose validity is rarely made explicit.

This work was motivated by a dynamical model of Bandura's social cognitive theory[15, 16], in which six psychosocial constructs that determine physical activity interact through 22 parameters (coupling strengths, input gains and time constants). When all 22 parameters were estimated from 31 days of step-count data, many stable parameter sets fitted the data with widely different quality (coefficients of determination between 4% and 86%), and local refinement from the best global estimate lowered the fit from 86% to 41%[17]. Such behaviour is typical when several parameters have nearly identical effects on the measured output, so that the data cannot separate them; in such models, a principled way to decide which parameters to estimate is indispensable.

Here we present a reduction method that works directly on the measured outputs defined in the PEtab format[18], including experimental conditions, measurement transformations and noise levels. It distinguishes biological parameters, which belong to the mechanistic model, from calibration parameters (scaling factors and offsets of the measurement process), which are always re-estimated, and selects only among

the biological ones. Candidates are ranked by sensitivity, redundant candidates are rejected by condition-number and variance-inflation tests, the reduced model is accepted only if re-estimating its parameters reproduces the full model under random parameter perturbations, and a final backward elimination removes every parameter that is not needed, so that no parameter of the final subset can be removed without losing admissibility. We evaluate the method on 34 benchmark problems with real data[8, 18], test the reduced models on held-out experiments, quantify their robustness to perturbation size and to the method's own settings, compare the selected subsets with those of four existing criteria under the same protocol, and report the computational cost of the complete method. The main mathematical statements on which the method relies are formally verified in the Lean 4 proof assistant with the Mathlib library[19, 20]; we state precisely what is and what is not proved.

Results

Overview of the method

Figure 1 summarises the procedure; all symbols are defined in the Methods and in Supplementary Table S2 (Supplementary Note *Notation of the method*). The input is a PEtab problem: an SBML model, the list of measured observables with their experimental conditions, measurement times, transformations (linear, logarithmic) and noise standard deviations σ_i, and the nominal parameter values θ_0 reported by the original authors. Each measurement i defines a scaled output $y_i(\theta) = h(g_i(x(t_i;\theta),\theta))/\sigma_i$, so that all outputs are comparable in units of measurement noise. Parameters are split into biological parameters θ_B (those of the SBML model) and calibration parameters θ_C (observation scalings and offsets); only biological parameters are candidates for selection, while calibration parameters are always re-estimated together with the selected ones.

The relative sensitivity matrix $J = \partial y/\partial \log\theta$ is computed once at θ_0 by central finite differences. In stage 1, biological parameters are added in decreasing order of their sensitivity energy $E_j = \|J_{:,j}\|^2$; a candidate is accepted only if the condition number κ and the largest variance inflation factor (VIF) of the normalised columns of J_S stay below a threshold, and is otherwise discarded permanently. Stage 1 stops when the accepted parameters capture a fraction τ_R of the total energy. The resulting subset S is then tested: in random scenarios where all parameters are perturbed, θ_S and θ_C are re-estimated with the remaining parameters fixed at θ_0, and S is admissible if the median relative fitting error e_{fit} is below τ_e and $|S| \geq 2$. The threshold $\tau_e = \sqrt{1-c^2}$ equals the smallest error attainable by rescaling an output change whose cosine similarity with the full change is c (Theorem 1); we use $c = 0.9$, so $\tau_e = 0.436$. If stage 1 does not yield an admissible subset, stage 2 repeats the selection with parameters ordered by their finite-perturbation effect s_j until an admissible subset is found. Finally, backward elimination removes any parameter whose removal keeps the subset admissible, so that no parameter of the result can be removed (Theorem 2).

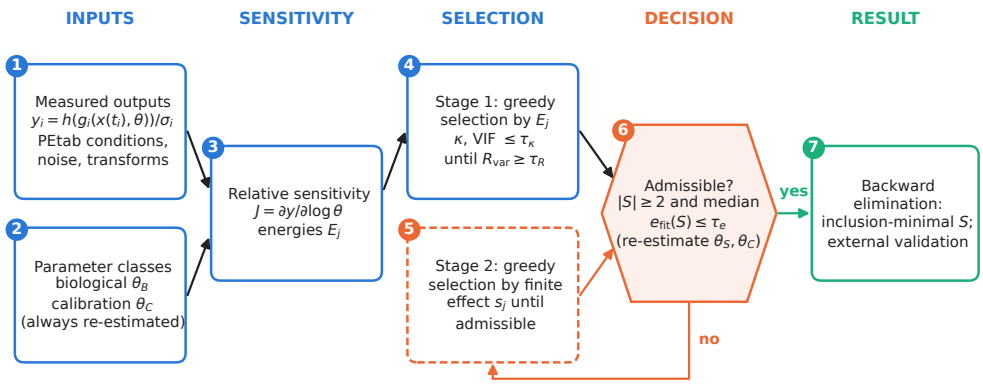


Fig. 1 Overview of the reduction method. Biological parameters are ranked by their effect on the measured outputs and selected with conditioning control (κ and VIF thresholds; stage 1, or stage 2 if needed, dashed). A subset is accepted when re-estimating it reproduces the full model; backward elimination then makes it minimal, and the reduced model is validated on held-out data.

A motivating case with exact collinearity

The social cognitive theory model[15–17] describes six psychosocial constructs through a linear system with 22 parameters and a single measured output (daily steps from one participant[17]). At the published estimate θ_0, the sensitivity matrix of behaviour is exactly rank-deficient (infinite condition number, maximum VIF 4.5×10^{15}), with nine pairs of parameters having cosine similarity above 0.95, making the full estimation problem ill-posed. The method selected two parameters (γ_{57} and β_{45}, the path from the external input through social satisfaction to behaviour), which, re-estimated on days 1–24, predicted the held-out week with $R^2 = 0.84$ (test χ^2/n 1.63), better than the full model refitted locally ($R^2 = 0.67$) or by differential evolution ($R^2 = 0.79$). Because θ_0 was estimated from all 31 days, this is an illustration anchored at θ_0, not an independent prospective validation; the model is not counted in the benchmark statistics below (Supplementary Note *The social cognitive theory model*, Supplementary Table S40 and Supplementary Fig. S15).

Benchmark: a few biological parameters suffice in most models

We applied the method to 34 PEtab benchmark problems with real measurements[8, 18] covering signalling, metabolism, gene regulation, cell cycle, immunology and epidemiology, with 3 to 294 parameters and 16 to 27,132 measurements (Table 1); Supplementary Note *System-by-system summary* describes each problem, the meaning of its selected parameters and its validation. One further benchmark problem (Froehlich, 4,231 parameters and 9,169 experimental conditions) was analysed only on its state trajectories because each evaluation of its measured outputs requires about 9,000 simulations; it is not counted here (see Computational cost).

The method found an admissible subset in 31 of the 34 problems (Fig. 2 and Table 1); in 25 of them admissibility was established by nonlinear re-estimation and

in six, the problems with the most expensive simulations (Alkan, Bachmann, Chen, Isensee, Lucarelli and Raimundez), by the first-order approximation of e_{fit} (Methods), so for these six it is not verified by nonlinear re-estimation. In 28 of them the subset contains only two biological parameters; two is also the minimum subset size enforced by the method ($k_{\min} = 2$), so this result does not imply that a single parameter would be insufficient—Supplementary Note *Single-parameter subsets* shows that one biological parameter already reaches the admissibility threshold in 15 of 27 tested problems. The remaining three subsets contain three (Zhao), five (Bruno) and ten (Lucarelli) biological parameters. Fifteen problems also have calibration parameters (up to 79 in Raimundez), which are re-estimated in every case. Counting both kinds, the reduced estimation problem has a median of 78% fewer free parameters than the full one (range 33–97%); among biological parameters alone, the median reduction is 88%. Backward elimination removed between one and five parameters in five problems (Supplementary Note *Selected subsets of the benchmark problems*, Supplementary Table S1). All 31 subsets satisfy the conditioning thresholds ($\tau_\kappa = \tau_{\mathrm{VIF}} = 10$; maximum observed $\kappa = 6.5$, VIF$= 9.8$), which bound the uniqueness and stability of the linearised least-squares estimates of the retained parameters (Supplementary Table S4, step 7) but do not require orthogonality; some selected pairs have cosine similarity near 0.9. Besides the admissibility condition $e_{\mathrm{fit}} \leq \tau_e$, which all of them meet, 17 subsets also have a median $\cos\Delta \geq 0.9$, so that a single rescaling of the reduced response already achieves the threshold (Theorem 1), and two of these (Alkan and Perelson) also keep at least 89% of the sensitivity energy, so that the first-order error bound applies as well; Supplementary Table S1 gives these levels of guarantee for every problem. A low R_{var} or $\cos\Delta$ does not prevent a good reduction, because the re-estimated parameters compensate for the effects of the fixed ones; the cosine criterion (Theorem 1) is sufficient but not necessary for admissibility (in Weber, with five calibration parameters, the median $\cos\Delta$ is 0.41 while $e_{\mathrm{fit}} = 0.04$; Supplementary Note *System-by-system summary*). Stage 1 alone was sufficient in seven problems; in the other 24, the energy ordering selected parameters whose re-estimation could not reproduce the full model, and the finite-effect ordering of stage 2 was needed. The three non-admissible problems fail for different reasons (Supplementary Notes *System-by-system summary* and *Random scenarios and large problems*). In Crauste, changes of 1% in single parameters make the solution of the T-cell model grow without bound within the measured horizon, so the sensitivity matrix and the admissibility test are not defined and no local reduction method is applicable. In Beer and Lang, the selection stopped at the conditioning limit with 18 and 13 parameters and $e_{\mathrm{fit}} = 50\%$ and 56%. In Lang this is a limitation of the greedy search rather than of the model: with $\tau_\kappa = \tau_{\mathrm{VIF}} = 5$ or 30 the search follows another path to admissible pairs, for example {`kDeE2f1`, `kSySkp2`} with $e_{\mathrm{fit}} = 13\%$; Lang is also the problem in which our simulation departs most from the benchmark reference.

Re-estimation is essential. When the selected parameters are simply set to their perturbed values, without re-estimation, the median relative error e_{rel} exceeds τ_e in 23 of the 31 admissible problems (Fig. 2b); after re-estimation the median error e_{fit} is 15% (maximum 43.5%). In other words, the selected parameters do not replicate the effect of every individual perturbation; rather, adjusting them compensates for

the perturbations of all other parameters in the measured outputs. This is precisely the property needed when the reduced model is fitted to data.

The selected parameters are also identifiable

A subset that reproduces the full model need not be determined by the data: a selected parameter could, for example, compensate for a measurement scaling factor, so that only their product is fixed. We therefore computed the profile likelihood of every selected parameter in the reduced problem, with $S \cup \Theta_C$ free, the other parameters at θ_0 and all measurements (Methods; Supplementary Note *Identifiability of the selected parameters*). In 25 of the 31 problems all selected parameters were practically identifiable (Table 1, column Ident.), that is, their 95% profile-likelihood interval was bounded on both sides within the range of the study (65 of the 74 selected parameters). The exceptions were Fujita, where a selected rate constant is confounded with a scaling factor of the measurement; Giordano and Perelson, whose recorded noise levels are far larger than the residuals, so that no parameter change is detectable; Okuonghae and Oliveira, in which one of the two parameters is identifiable and the profile of the other could not be simulated; and Smith, whose optimum for one parameter lies at the bound of the study. With all parameters of the full model free, in contrast, only 20 of 52 selected parameters were identifiable (Supplementary Note *Profile likelihood*): the data do not determine all parameters of these models, but, once the others are fixed at their published values, they determine the few that the method selects. Which parameters are identifiable is predicted, in every problem whose profiles could be evaluated, by the profiled information $I_j = 1/[(A^\top A)^{-1}]_{jj}$, where A is the sensitivity of the noise-scaled residuals with respect to the free parameters; this is the quantity on which a sufficient condition for practical identifiability, verified in Lean, is based (Methods). Because only the selected and calibration parameters are free, this check is inexpensive: determining the identifiability of the selected parameters took a median of 34 s of computing time per problem (less than ten minutes in 24 of the 31 problems), and it was also inexpensive in the large problems, in which the profile likelihood of the full model took hours or was not run.

Prediction of held-out experiments

Reproducing the full model is not the same as predicting new data. We therefore split the measurements of each problem into training and test sets, by experimental condition when several conditions were available (21 problems) and by time otherwise, using the later time points as test set (10 problems); calibration or condition-specific parameters that appear only in the test set were excluded. The reduced model (re-estimating θ_S and θ_C) and the full model (re-estimating all parameters) were fitted to the training data starting from θ_0 and evaluated on the test data. In two problems a split was impossible (Bertozzi, whose selected parameters only act in the test conditions; Blasi, with a single condition and a single time point).

With the non-selected parameters kept at their published values θ_0, the reduced model predicted the held-out experiments at least as accurately as the full model

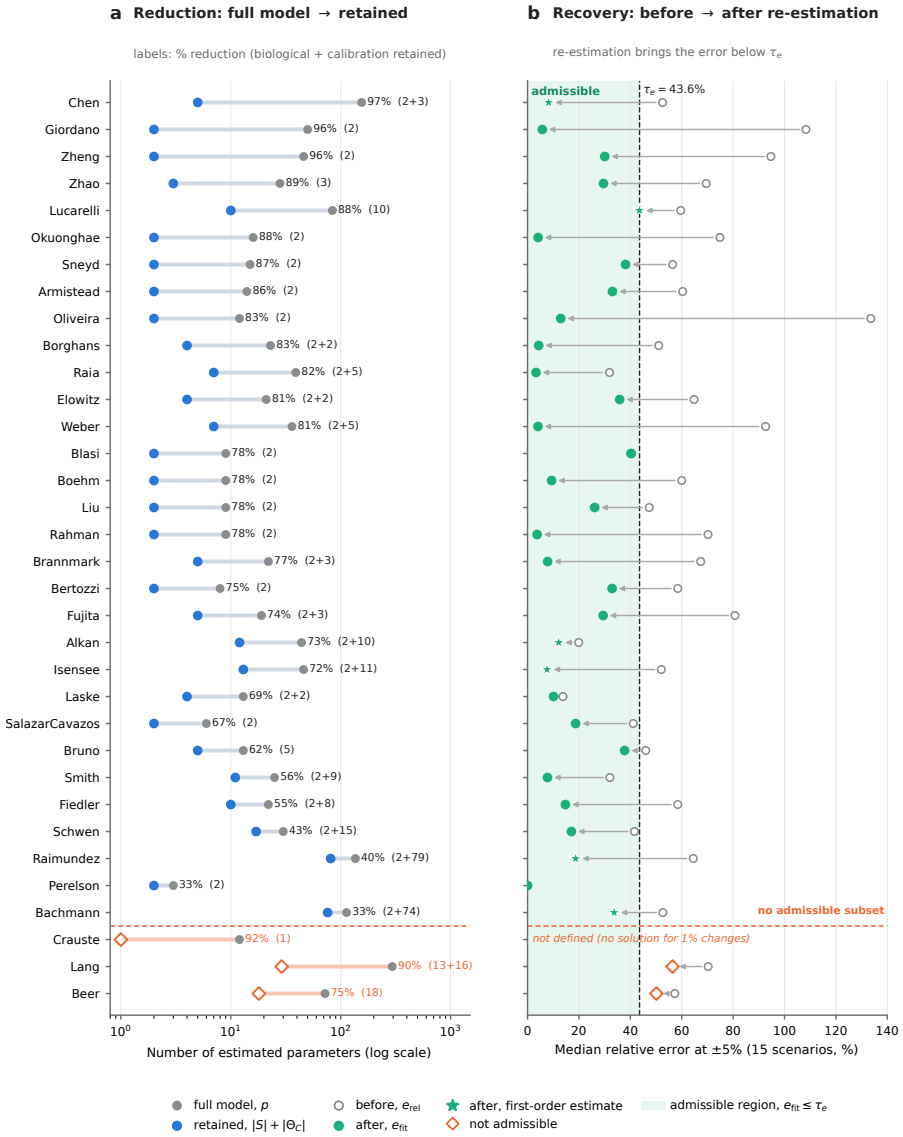


Fig. 2 Parameter reduction and recovery of the full-model output in 34 benchmark problems. Problems are ordered by the reduction achieved; the two without an admissible subset are at the bottom. **a**, Estimated parameters of the full model (p, grey) and parameters re-estimated in the reduced model (selected biological plus calibration parameters, blue). **b**, Median relative error over 15 random scenarios at ±5% before (e_{rel}, open circles) and after re-estimating the retained parameters (e_{fit}, green; stars, first-order estimate). The shaded region is the admissible region, $e_{\mathrm{fit}} \le \tau_e = 43.6\%$.

Table 1 Per-problem results. p, number of estimated parameters; m, number of measurements; $|\Theta_C|$, calibration parameters (always re-estimated); $|S|$, selected biological parameters; Stage, stage that produced the subset; e_{rel} and e_{fit}, median relative error over 15 scenarios at ±5% without and with re-estimation (admissibility: $e_{\text{fit}} \leq 43.6\%$; italic, not admissible; *, e_{fit} computed to first order because of simulation cost; †, not defined because the model has no solution on the measured horizon for 1% parameter changes); Pattern, behaviour of the median e_{fit} for perturbation sizes from ±1% to ±50% (A, admissible at all sizes; B, admissible up to at least ±10%, first failing at ±20% or more and not recovering; C, irregular; D, not admissible at ±10% or less; –, not applicable); Level, level of guarantee of the admissible subset (1, $e_{\text{fit}} \leq \tau_e$; 2, in addition median $\cos\Delta \geq c$; 3, in addition $R_{\text{var}} \geq \tau_R$; Methods); Ident., selected parameters whose profile likelihood in the reduced problem exceeds $\Delta\chi^2 = 3.84$ on both sides within the bounds, out of $|S|$, and in parentheses those for which the hypotheses of the Lean identifiability theorem hold (Table 3); Test χ^2/n, mean squared noise-scaled residual on held-out experiments for the reduced and the full model, both re-estimated on the training experiments (values near 1 mean errors of the size of the measurement noise; –, not evaluable or not run).

System	p	m	$\lvert\Theta_C\rvert$	$\lvert S\rvert$	Stage	e_{rel} (%)	e_{fit} (%)	Level	Pattern	Ident. (Lean)	Test χ^2/n red. / full
Alkan	44	1733	10	2	1	20	12.1*	3	A	2/2 (1)	2.75 / 3.37
Armistead	14	58	0	2	2	60	33.0	1	B	2/2 (2)	1.22 / 1.37
Bachmann	113	541	74	2	2	53	33.7*	1	A	2/2 (0)	4.77 / 7.35
Beer	72	27132	0	18	2	57	*50.1**	–	–	–	–
Bertozzi	8	22	0	2	2	58	32.9	2	B	2/2 (0)	–
Blasi	9	252	0	2	1	41	40.2	2	B	2/2 (2)	–
Boehm	9	48	0	2	2	60	9.4	2	A	2/2 (2)	2.57 / 11.22
Borghans	23	111	2	2	2	51	4.3	1	A	2/2 (0)	0.74 / 0.74
Brannmark	22	43	3	2	2	67	7.8	1	A	2/2 (0)	5.96 / 6.28e+03
Bruno	13	77	0	5	1	46	37.8	2	A	5/5 (5)	0.91 / 208
Chen	155	120	3	2	2	53	8.2*	1	A	2/2 (0)	20.88 / 17.71
Crauste	12	21	0	1	2	–	*n.d.*†	–	–	–	–
Elowitz	21	58	2	2	1	65	35.8	1	B	2/2 (0)	1.50 / 3.89
Fiedler	22	72	8	2	2	58	14.7	1	A	2/2 (0)	11.21 / 9.32
Fujita	19	144	3	2	2	81	29.4	1	A	0/2 (0)	5.94 / 8.03
Giordano	50	313	0	2	2	108	5.8	2	A	0/2 (0)	<0.01 / <0.01
Isensee	46	687	11	2	1	52	7.6*	1	A	2/2 (0)	1.08 / 1.08
Lang	294	9600	16	13	2	70	*56.4**	–	–	–	–
Laske	13	42	2	2	1	14	10.1	2	A	2/2 (1)	0.76 / 1.87
Liu	9	100	0	2	2	47	26.1	2	A	2/2 (0)	0.95 / 1.02
Lucarelli	84	1755	0	10	2	60	43.5*	1	C	10/10 (4)	1.66 / 3.34
Okuonghae	16	92	0	2	2	75	4.1	2	A	1/2 (0)	4.67e4 / 1.22e5
Oliveira	12	120	0	2	2	134	12.9	2	A	1/2 (0)	912 / 8.13e4
Perelson	3	16	0	2	1	0	0.0	3	A	0/2 (0)	<0.01 / <0.01
Rahman	9	23	0	2	2	70	3.7	2	A	2/2 (2)	<0.01 / 0.03
Raia	39	205	5	2	2	32	3.3	2	A	2/2 (0)	8.22 / 5.10
Raimundez	136	627	79	2	2	64	18.7*	1	A	2/2 (0)	1.17 / 1.43
SalazarCavazos	6	18	0	2	2	41	18.7	2	A	2/2 (0)	102 / 124
Schwen	30	286	15	2	2	42	17.1	2	A	2/2 (0)	8.20 / 8.40
Smith	25	62	9	2	2	32	7.8	2	A	1/2 (0)	1.50e4 / 1.50e4
Sneyd	15	135	0	2	2	56	38.1	1	A	2/2 (2)	2.22 / 2.56
Weber	36	135	5	2	2	93	4.1	1	A	2/2 (0)	1.00 / 15.95
Zhao	28	82	0	3	2	70	29.6	2	A	3/3 (1)	0.35 / 0.35
Zheng	46	60	0	2	2	95	30.0	1	A	2/2 (2)	1.58 / 20.82

re-estimated on the same training data (test χ^2/n within 5% of the full model or lower) in 26 of the 29 problems in which both could be compared (Fig. 3a; the three problems with noise-free data are not shown). The exceptions were Raia (test χ^2/n of 8.2 versus 5.1), Fiedler (11.2 versus 9.3) and Chen (20.9 versus 17.7, both far above the noise level). In several problems the full model overfitted the training data and predicted much worse, for example Brannmark (test χ^2/n 6.0 versus 6,277), Bruno (0.9 versus 208), Weber (1.0 versus 16.0) and Zheng (1.6 versus 20.8). The re-estimation of the full model in these over-parameterised problems can end in different local optima; all real-data numbers in this paper come from a single run. In five problems even θ_0 has a test $\chi^2/n > 10$ (Chen, Okuonghae, Oliveira, Salazar-Cavazos and Smith). In Chen, Salazar-Cavazos and Smith our simulation agrees with the reference simulation of the benchmark (Supplementary Note *Verification of the simulator*), so a simulation error is not the cause there; the benchmark provides no reference simulation for Okuonghae and Oliveira, so for these two we cannot exclude it. A large χ^2/n can reflect recorded noise levels that understate the scatter of the data, a misspecified model or inadequate parameter values, and our analysis does not distinguish between these causes.

This comparison favours the reduced model in one respect. The published estimate θ_0 was obtained from all data, including the test experiments, and the reduced model inherits that information through the parameters it keeps fixed. Two checks show how much this matters (Fig. 3c; Methods). First, θ_0 alone, without any re-estimation, predicted the test data better than the reduced model, by more than 5%, in 20 of 29 problems (median ratio of test χ^2/n 0.90). Second, in a nested protocol the non-selected parameters were fixed at a full-model fit to the training data instead of at θ_0, which removes the leakage through the fixed values; this nested reduced model predicted worse than the θ_0-anchored one, by more than 5%, in 21 of 29 problems (median ratio 1.35). The better prediction of the reduced model on real data therefore reflects the value of anchoring at published estimates, not a property of the selected subset alone.

To measure prediction without this confound we generated synthetic data from a known truth θ^* that differs from θ_0 by up to $\pm 20\%$ in every estimable parameter, with Gaussian noise of the recorded size, so that θ_0 has not seen the test data (it still indicates the region in which the truth lies, so the test is local), and repeated the experiment with 20 random seeds (Supplementary Note *Synthetic prediction experiment*). Among the cases in which the parameter change exceeded the noise, the reduced model reached a relative prediction error $e_{\mathrm{pred}} \leq \tau_e$ in 56% (178 of 320; 31–76% depending on the seed) and predicted at least as well as the full model in 57% (160 of 282; 40–73%). The outcome was stable within each problem and divided the benchmark into two groups (Fig. 3b): the reduced model predicted better in almost every seed of some problems (for example Weber 20 of 20 seeds, Zheng 18 of 20, Brannmark 17 of 18, Isensee 12 of 12), in which the full model overfits the training data, and worse in almost every seed of others (Oliveira 0 of 20, Fujita 1 of 18, Okuonghae 2 of 20). On data that θ_0 has not seen, the two-parameter models thus predict neither better nor worse than the full models overall (equivalence was not tested), with a consistent advantage when the full model overfits. The advantage

depends on how far the truth is from θ_0: repeating the experiment with truths within $\pm 5\%$ and $\pm 50\%$ of θ_0 (ten seeds), the reduced model reached $e_{\text{pred}} \leq \tau_e$ in 64% and 51% of the cases with signal (54% at $\pm 20\%$ on the same seeds) and predicted at least as well as the full model in 67% and 47%. The success also grows with the size of the true change in noise units (Methods): 1%, 24%, 44% and 75% of the cases at $\pm 20\%$ when it is below 0.3σ, between 0.3σ and σ, between σ and 3σ, and above 3σ per measurement (Supplementary Note *Synthetic prediction experiment*).

Comparison with existing selection criteria

We compared the subsets of our method with those of four existing criteria that also select which parameters to estimate: pivoted QR of the normalised sensitivity matrix[32]; the collinearity index of Brun *et al.*[14] (the most important parameters with collinearity index at most 10 and the largest determinant measure); Morris screening of elementary effects over the whole range $[\theta_0/10,\ 10\theta_0]$[12]; and profile likelihood[5, 29], retaining the parameters with the narrowest profile-likelihood confidence intervals. To compare the criteria rather than the subset sizes, every criterion chose the same number $k = |S|$ of biological parameters as our method, from the same training data, and every subset was evaluated with the same three tests: prediction of the held-out experiments, reproduction of the full model (the admissibility test of our method) and prediction of synthetic data (Methods; Table 2 and Fig. 4). Two further baselines re-estimated only the calibration parameters ($k = 0$) or k biological parameters drawn at random (median of 20 draws). Profile likelihood, which re-estimates the full model at every profile point, was not run for Chen and Raimundez, the two problems with the most expensive simulations and with 155 and 126 free parameters, and Chen was also excluded from the Brun and Morris comparisons because of its simulation cost (Supplementary Notes *Pivoted QR selection*, *Profile likelihood* and *Brun and Morris selections*).

Profile likelihood gives an independent view of how many parameters the data determine. Over the 28 problems in which it was computed, the median number of biological parameters whose profile exceeded the 95% threshold on both sides was 2.5 (17% of the biological parameters), six problems had none, and 14 had two or fewer. This agrees with the size of the subsets found by our method, although the two criteria rarely chose the same parameters (none or one in common in most problems): a flat profile means that other free parameters can compensate for changes in a parameter, whereas our method asks which parameters suffice when all others are fixed at θ_0.

On real data, the criteria were practically indistinguishable (Fig. 4c). Pairwise Wilcoxon tests over problems found no significant difference between our subsets and those of Brun ($p = 0.27$), Morris ($p = 0.46$), profile likelihood ($p = 0.44$) or random subsets ($p = 0.27$); pivoted QR and calibration-only re-estimation were slightly better (median ratio of test χ^2/n 0.99 in both; $p = 0.03$ and 0.003). Only in Oliveira did the specific choice matter, with our subset predicting 22 to 500 times better than the random and calibration-only baselines (Supplementary Note *Comparison on real data*). As expected for sloppy models[10], many different small subsets fit the training data and predict similarly when the remaining parameters are kept at θ_0, so this test does not discriminate between criteria.

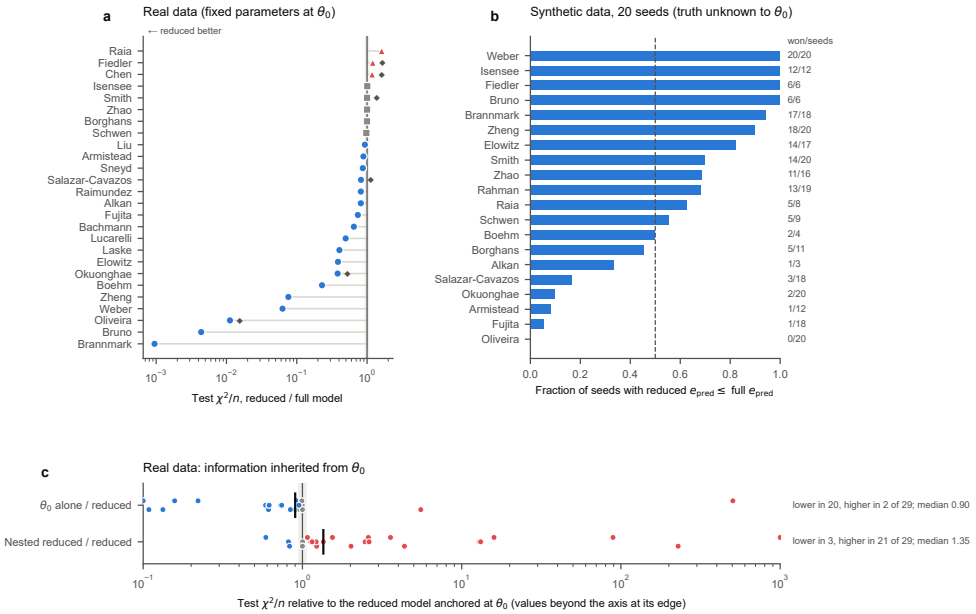


Fig. 3 Prediction of held-out experiments by the reduced and the full model. a, Real data: ratio of the test χ^2/n (mean squared noise-scaled residual on the held-out experiments) of the reduced model to that of the full model re-estimated on the same training data, on a logarithmic scale ($n = 26$ problems), with the non-selected parameters kept at θ_0. Left of the vertical line (circles) the reduced model predicts better; the grey band marks equivalence within 5% (squares); triangles mark worse predictions; ♦ flags a reduced test χ^2/n above 10, where a favourable ratio does not imply a good absolute fit. **b**, Synthetic data from a known truth, which θ_0 has not seen: for each problem, fraction of the 20 seeds in which the reduced model predicted at least as well as the full model (smaller e_{pred}), among the seeds in which the parameter change exceeded the noise (seeds won / seeds with signal on the right; problems with fewer than three such seeds are not shown). Problems are ordered by this fraction. **c**, Real data, information inherited from θ_0: test χ^2/n of the published estimate θ_0 without re-estimation, and of the nested reduced model (non-selected parameters fixed at a full-model fit to the training data instead of θ_0), each relative to the reduced model anchored at θ_0; one point per problem ($n = 29$), vertical bar at the median; blue, lower by more than 5%; red, higher by more than 5%; grey band, within 5%. Problems without a training/test split (Bertozzi, Blasi), without a full re-estimation (in **b**, Chen and Raimundez) or with noise-free data (**a**: Giordano, Perelson, Rahman) are not shown. Per-problem values and scatter plots are given in Supplementary Notes *Prediction of held-out experiments* and *Leakage-free nested validation.*

The criteria differed clearly in whether their subsets reproduced the full model (Fig. 4a). At the perturbation size used by our method (±5%), our subsets were admissible in all 28 comparable problems, against 26 of 28 for Brun, 23 of 28 for Morris, 18 of 27 for profile likelihood and 14 of 28 for pivoted QR, and there was no problem in which another criterion produced an admissible subset and ours did not (Supplementary Note *Admissibility of the subsets of other criteria*). In the two largest problems in which profile likelihood was run, its subset was admissible in Bachmann (median e_{fit} 0.30, sharing one of its two parameters with ours) but not in Lucarelli (0.74), where our subset reached 0.435 (Supplementary Note *Profile likelihood*). Because our method selects subsets precisely by this test, we repeated it at ±20%, a level used by none of the criteria: our subsets remained admissible in 27 of 28 problems, against 25 of 28, 23 of 28, 18 of 27 and 15 of 28 for the other criteria (exact McNemar test over problems, Holm-adjusted: $p = 0.5$ against Brun, 0.24 against Morris, 0.02 against profile likelihood and 0.002 against pivoted QR). The same directions used at ±5% were rescaled to ±20%, so this is a test of the size of the perturbation, not an independent sample of directions. The same difference appeared in prediction when θ_0 has not seen the test data (Fig. 4b): over 282 synthetic cases in 24 problems, our subsets reached $e_{\mathrm{pred}} \leq \tau_e$ in 56% of cases, Brun's in 55%, Morris's in 48%, those of profile likelihood in 39% and those of pivoted QR in 34%; resampling whole problems, the advantage was 22 percentage points over pivoted QR (95% confidence interval 10–34; Holm-adjusted $p < 0.001$), 17 over profile likelihood (8–28; $p = 0.002$) and 9 over Morris (2–16; $p = 0.008$), whereas the difference from Brun (+1; −1 to +4; $p = 0.33$) was not significant. The median prediction error itself did not differ significantly between criteria (Supplementary Note *Comparison of selection criteria on synthetic data*). The distinguishing property of our method is therefore not a lower average prediction error but reliability: its subsets are verified to reproduce the full model, and in the synthetic test they also reached the prediction threshold more often than those of pivoted QR, profile likelihood and Morris (a difference that we observed, not one that the verification guarantees); Brun's collinearity index, which shares the sensitivity-based ranking, comes closest.

Robustness to the size of parameter changes

Admissibility is decided with perturbations of ±5%. To determine how far the conclusion extends, we kept each final subset fixed and repeated the evaluation with perturbations of ±1, 5, 10, 20, 30, 40 and 50% (15 scenarios per level). The scenarios are paired: the same fifteen random directions $u^{(k)}$ are used at every level, $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, so that differences between levels are due only to the size ε, and the ±5% level reproduces the scenarios of the selection. We classified the behaviour of the median e_{fit} into four patterns: A, admissible at all levels; B, admissible up to at least ±10%, first failing at ±20% or more and not recovering; C, irregular (failing and then passing again); D, failing at ±10% or less.

All 31 admissible problems were evaluated. Twenty-six follow pattern A (Fig. 5): their median e_{fit} stays below the threshold up to ±50%, typically varying by only a few percentage points across levels. Four follow pattern B: Armistead, Bertozzi, Blasi and Elowitz, whose median errors are already between 33% and 40% at ±5% and

Table 2 Comparison of selection criteria. All criteria choose the same number $k = |S|$ of biological parameters as our method and are evaluated with the same protocol. Admissible: subsets whose re-estimation reproduces the full model (median $e_{\text{fit}} \leq \tau_e$) for parameter changes of ±5% and ±20%. Synthetic: fraction of the 282 common synthetic cases with signal (24 problems, 20 seeds) in which $e_{\text{pred}} \leq \tau_e$, and difference from our method with a 95% confidence interval from resampling problems. Real data: median ratio of the test χ^2/n to that of our method (below 1, better than ours; non-selected parameters at θ_0) and Wilcoxon p-value over problems (unadjusted; exploratory comparison). Admissibility is compared in 28 problems (Bertozzi excluded, see Methods); profile likelihood in 27 of them (not run for Raimundez, Methods). Per-problem values are given in the Supplementary Notes on the comparison of selection criteria.

		Admissible		Synthetic: $e_{\text{pred}} \leq \tau_e$		Real data
Criterion	Basis	±5%	±20%	Rate	Ours − criterion (95% CI)	χ^2 ratio (p)
Our method	Sensitivity, κ/VIF, re-estimation test, backward elimination	**28/28**	**27/28**	56%	–	–
Brun	Sensitivity and collinearity index	26/28	25/28	55%	+1 (−1 to +4)	1.00 (0.27)
Morris	Global elementary effects	23/28	23/28	48%	+9 (+2 to +16)	1.00 (0.46)
Profile likelihood	Repeated re-estimation of the full model	18/27	18/27	39%	+17 (+8 to +28)	1.00 (0.44)
Pivoted QR	Linear independence of sensitivities	14/28	15/28	34%	+22 (+10 to +34)	0.99 (0.03)
Random	k parameters at random (median of 20)	–	–	–	–	1.00 (0.27)
Calibration only	No biological parameter ($k = 0$)	–	–	–	–	0.99 (0.003)

grow until they cross the threshold, between ±10% and ±20% (Elowitz), ±20% and ±30% (Bertozzi, Blasi) or ±30% and ±40% (Armistead). Lucarelli is the only case of pattern C: its median error stays between 43% and 47% at all levels, that is, on the threshold rather than clearly above or below it. No problem follows pattern D. The median does not show how many individual scenarios are reproduced, so we also report the pass rate, the fraction of the valid scenarios (those that could be simulated and whose output change exceeds the numerical noise floor) with $e_{\text{fit}} \leq \tau_e$ (Fig. 5b). All fifteen scenarios were valid in 28 of the 31 problems; in Chen, Isensee and Perelson only 8, 4 and 2 were. At ±5%, 13 of the 31 problems reproduce the full model in every valid scenario (10 of them with all fifteen valid) and the median pass rate is 93%; the other 18 have at least one scenario above the threshold. The same fifteen directions are rescaled at every perturbation size, so the levels test the size of the change, not new directions; with independent directions at each size (Methods), 24

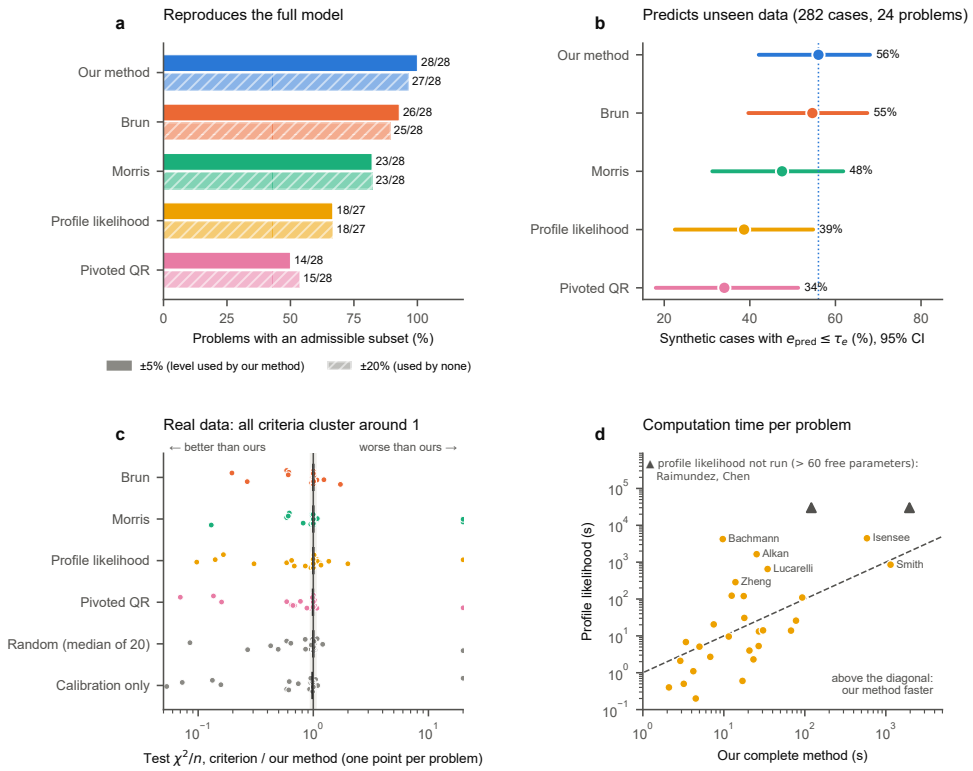


Fig. 4 Our method compared with four existing selection criteria. a, Number of problems in which the subset of each criterion is admissible, i.e. its re-estimation reproduces the full model, for parameter changes of ±5% (the level used by our method) and ±20% (used by none); 28 problems (27 for profile likelihood). **b**, Fraction of synthetic cases in which the subset predicts the truth within τ_e (282 common cases, 24 problems, 20 seeds), with 95% confidence intervals from resampling problems. **c**, Real data: ratio of the test χ^2/n of each criterion to that of our method, one point per problem, vertical bar at the median (log scale; grey band, ±5%; values outside the axis are drawn at its edge); all criteria cluster around 1. **d**, Computation time of our complete method and of profile likelihood per problem (log scales); points above the diagonal are faster with our method; problems in which profile likelihood was not run (Chen and Raimundez) are shown at the top.

problems follow pattern A, one pattern B and six pattern C (against 26, 4 and 1 with the same directions); 22 problems keep pattern A in both designs, and the nine that change are those whose median error is close to the threshold at large perturbations (Supplementary Note *Robustness to the size of parameter changes*). Among the 26 problems of pattern A, 9 pass every scenario at every level, whereas 7 (Borghans, Bruno, Fujita, Liu, Sneyd, Zhao, Zheng) fall to a pass rate of 60–73% at some level, so their median is below the threshold while four to six of the fifteen scenarios are not reproduced. The reductions therefore remain valid well beyond the ±5% used for selection in most problems, whereas problems that are admissible only with a small margin at ±5% are those that lose admissibility for large changes. The full distribution of the error for every problem and level (interquartile range, 5–95% range,

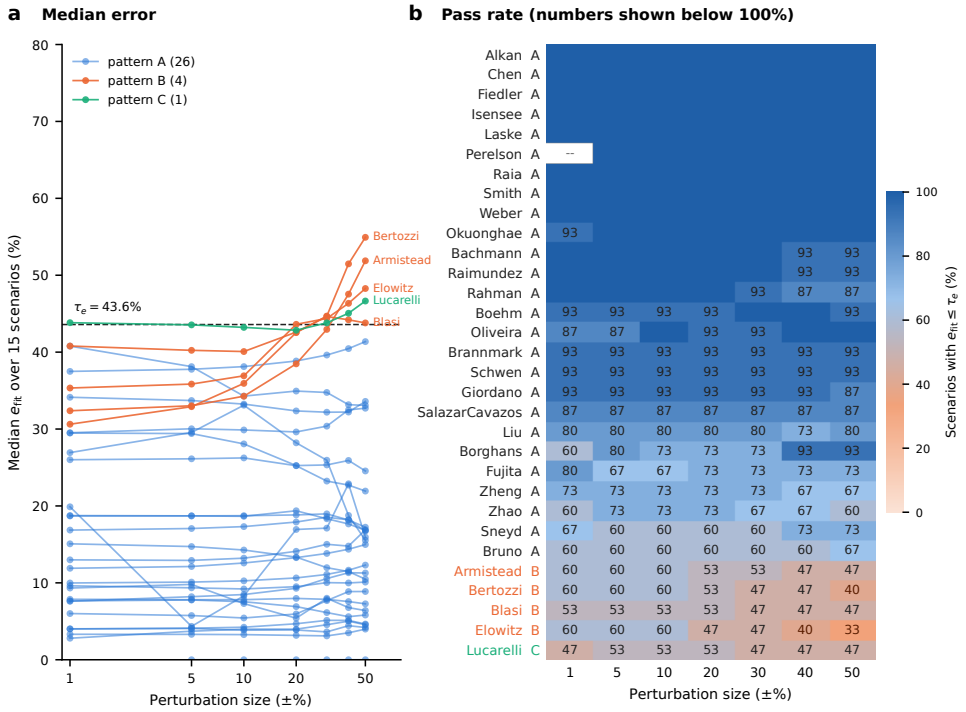


Fig. 5 Robustness of the selected subsets to the size of parameter changes. a, Median relative error after re-estimation (e_{fit}) over 15 random scenarios per level as a function of the perturbation size, for the final subset of each of the 31 admissible problems, coloured by robustness pattern (A, admissible at all levels; B, first failing at ±20% or more; C, irregular); dashed line, $\tau_e = 43.6\%$. **b**, Pass rate: percentage of the scenarios with $e_{\mathrm{fit}} \leq \tau_e$, with the value written when it is below 100%. The same fifteen random directions are used at every level. –, no valid scenario (Perelson at ±1%, where the output change is below the numerical noise floor).

maximum and pass rate) is given in Supplementary Note *Robustness to the size of parameter changes* (Supplementary Tables S11 and S12 and Supplementary Fig. S4).

Sensitivity to method settings and computational cost

The method depends on a finite-difference step δ, integrator tolerances, the perturbation size used for admissibility and four thresholds (τ_R, τ_κ, τ_{VIF}, τ_e). We re-ran the complete method on the 26 small and medium problems changing one setting at a time (Methods; Supplementary Note *Sensitivity to the method settings*); in the baseline, 25 of them are admissible. With $\delta = 0.1\%$ or 5%, with either alternative set of integrator tolerances and with τ_R between 0.80 and 0.95, all 25 subsets were unchanged. With perturbations of ±1% or ±10% the subset changed only in Zhao, and admissibility only for Perelson at ±1%, where the perturbed output change falls below the numerical noise floor. A conditioning threshold of 5 or 30 changed three or four subsets without changing the number of admissible problems.

The error threshold has the largest influence: with $\tau_e = 0.10$, 0.20, 0.30 and 0.50, 10, 15, 18 and 23 subsets were unchanged and 22, 24, 25 and 25 problems admissible; $\tau_e = 0.10$ requires a mean of 3.7 biological parameters, whereas τ_e between 0.30 and 0.50 gives 2.1–2.5. The threshold therefore acts as an accuracy–parsimony trade-off; it is set through the cosine requirement $c = 0.9$ by $\tau_e = \sqrt{1 - c^2}$ (Theorem 1), so the only choice is c, whose effect is covered by the scan. The value $c = 0.9$ is a design choice, not a tolerance validated for any biological application: e_{fit} is measured relative to the change of the model response caused by the perturbation, not relative to the data. To relate it to the precision of the measurements, we also expressed the remaining error in units of the recorded noise (Supplementary Table S33): at ±5%, its median root mean square per measurement was below 0.1σ in 17 of 31 problems and below σ in 29; it exceeded the noise only in Okuonghae and Oliveira, in which a ±5% perturbation changes the outputs by about 30 and 120 noise units. An application that needs a closer reproduction should choose c from the accuracy it requires, at the price in subset size shown by the scan. The minimum subset size $k_{\min} = 2$ is a design choice: a single biological parameter, re-estimated with the calibration parameters, already reached the threshold in 15 of 27 problems examined; in the synthetic prediction experiment the selected subset predicted better than the best single parameter in 9 of these 15, but on real data the single parameter had the lower test χ^2/n in 10 of 15 (Supplementary Note *Single-parameter subsets*). The admissibility test uses the median over $N = 15$ random scenarios rather than requiring all of them to pass, because individual scenarios near the boundary of the parameter domain can fail due to ODE nonlinearity; six problems pass with 8–9 of the 15 scenarios. For Lucarelli, the first-order median e_{fit} (0.435) lies within 10^{-3} of τ_e, so its admissibility is borderline, although it is stable across perturbation sizes (43–47% from ±1% to ±50%). The random seed and the number of scenarios N were not varied.

The cost of the selection is dominated by the computation of J ($2p$ simulations) and of the stage-2 ordering (p simulations), so the number of sensitivity simulations grows linearly with the number of parameters; all of them are independent and can run in parallel, and the selection, condition-number tests and first-order e_{fit} only use linear algebra on J. The admissibility test adds 15 scenarios with re-estimation for each candidate subset, a cost that depends on the simulation time and on the number of retained parameters rather than on p. Run from scratch, the complete method (selection, admissibility and backward elimination) took from a few seconds to 33 minutes per problem (median 19 s; Chen, with 35 s per simulation, was the slowest), on a shared multi-core server (Supplementary Note *Computational cost*). Profile likelihood, which requires repeated re-estimation of the full model, was much slower on problems with expensive simulations (Alkan 27 min against 25 s for our method; Zheng 5 min against 14 s; Isensee more than one hour against 10 min). Over the 28 problems in which both could be run, our method was faster in 12 (Supplementary Table S35); on small problems, profile likelihood was as fast as or faster than our method, because the admissibility test has a fixed cost. The advantage of the method is therefore not speed on small problems but that its cost grows slowly with the size of the problem. For the two admissible problems with more than 60 free parameters in which profile likelihood was run, Bachmann (90)

and Lucarelli (72), it took 70 and 11 min with the profile points distributed over 60 cores (40 and 9 h of computing time), against 10 and 35 s for our method (Fig. 4d); for Chen and Raimundez, whose simulations are the most expensive, it was not run, whereas our method processed them in 33 and 2 min. The method also ran on Lang, with 294 parameters, in about 3 min, although there it found no admissible subset. To show what selecting and verifying a subset costs, and where the computing time of the whole study went, Supplementary Note *Computing time of the analyses* gives the time of every analysis for every problem. Selecting a subset took less than ten minutes in 32 of the 34 problems and determining the identifiability of its parameters a median of 34 s, feasible in all 31 admissible problems; all analyses of the study together took about 169 h of computing time, 86% of it in five problems with slow simulations or many calibration parameters (Isensee, Raimundez, Chen, Smith and Lang), and the median over the problems of all analyses together was 21 min. On state trajectories, the 4,231-parameter Froehlich model required 12,693 simulations and 2.6 h for the sensitivities; for models of this size on measured outputs, forward sensitivity equations[21] compute J in a single integration of an augmented system, whose cost grows with the number of parameters; adjoint methods are efficient for gradients of a scalar objective but not for the full output Jacobian needed here. The linear growth refers to the number of sensitivity simulations; the total cost also depends on the number of candidate subsets tested, on the number of calibration parameters and on the simulation time, and the times reported are observed wall-clock times on a shared server.

Formal verification of the mathematical foundations

The mathematical statements on which the method relies and that are verified in Lean 4 with Mathlib are summarised in Supplementary Table S4; the complete development comprises 32 modules with 304 theorems and lemmas and 28 verified benchmark models; Supplementary Tables S6 and S7 give the notation of the formalisation and the statement of every theorem and lemma, and all proofs compile together with a single command (Code availability). None of the proofs uses unproved assumptions (`sorry`) or additional axioms. Table 3 summarises, for each step of the method, what is proved, when the statement applies to a numerical result and to which results of this study it applies. Two statements are central to the method:

Theorem 1. *For vectors $d \neq 0$ and $b \neq 0$ in a real inner-product space,* $\min_{\lambda \in \mathbb{R}} \|d - \lambda b\| / \|d\| = \sqrt{1 - \cos^2(d, b)}$.

Theorem 2. *Let $\mathcal{A}$ be any admissibility predicate on finite sets and S an admissible set. Backward elimination (removing one element at a time whenever the remaining set is still admissible, until no element can be removed) returns an admissible subset of S from which no element can be removed without losing admissibility. If, in addition, $\mathcal{A}$ is monotone (every superset of an admissible set is admissible), no proper subset of the result is admissible.*

Theorem 1 relates the threshold τ_e to the cosine: whenever $\cos\Delta \geq c$, rescaling the reduced response already gives a relative error of at most $\sqrt{1-c^2}$, and re-estimating several parameters can only do better to first order. The converse does not hold, and admissibility itself is decided by the numerically computed e_{fit}. Theorem 2 gives the precise sense in which the selected subsets are minimal: no parameter can be removed. Our admissibility criterion is monotone when the re-estimation reaches its optimum, because re-estimating more parameters never increases the fitting error (also verified), so in that case no proper subset of the result is admissible; with a local optimiser this holds only approximately. Minimality is relative to the criterion, which includes the design choice $|S| \geq 2$, and is not minimal cardinality: a different, smaller admissible subset that is not contained in the result may exist. The other verified statements cover the remaining steps: re-estimating more parameters never increases the fitting error and $e_{\mathrm{fit}} \leq e_{\mathrm{rel}}$; the selection keeps κ and VIF below their thresholds and permanently discarding a rejected candidate is correct because both quantities can only increase when parameters are added; κ and VIF bounds imply that the linearised least-squares problem for the retained parameters has a unique and stable solution (which is not a proof of identifiability of the nonlinear model); error bounds on the states transfer to the measured outputs, including logarithmic transformations; and the first-order version of e_{fit} used for the largest models differs from the exact one by a term of second order in the absolute error (which does not by itself bound the relative error used in the decision). A further group of verified statements concerns the identifiability of the selected parameters (step 32 of Supplementary Table S4). Theorem `practically_identifiable_grad` gives a sufficient condition for a parameter of the reduced problem to be practically identifiable, that is, for its 95% profile-likelihood interval to be bounded on both sides within the range of the study: the guaranteed half-width t^*, computed from the profiled information I_j, the residual gradient γ at the optimum and a bound ε on the nonlinear remainder, must be smaller than the distance to both bounds (Methods). Applied to the benchmark, its linear version ($\gamma = \varepsilon = 0$, the Wald half-width) agreed with the profile classification for all 72 selected parameters whose profile could be evaluated, holding for the 65 identifiable parameters and failing for the other seven; with the remainder evaluated at the computed profile points, the full hypotheses held for 24 parameters and never for a parameter whose profile is not identifiable. In seven problems (Armistead, Blasi, Boehm, Bruno, Rahman, Sneyd and Zheng), every selected parameter satisfies these hypotheses and the model itself is verified in Lean, so their subsets are at the same time admissible and covered by the identifiability theorem; in the other 24 admissible problems, identifiability rests on the computed profiles, and the theorem is not applicable because the nonlinearity of the model is too large over the range of the study (Supplementary Note *Identifiability of the selected parameters*). The verification is local: it does not prove that a subset remains adequate far from θ_0, which is assessed empirically above. The theorems are implications; the quantities to which they are applied (κ, VIF, R_{var}, e_{fit}) are computed in floating point, and Supplementary Table S5 states the scope of each guarantee and the numerical values that connect it with our results. The model files used by Lean were checked against the SBML models of the benchmark, with

Table 3 What the formal verification guarantees. For each step of the method, the modules of the Lean development (numbers of Supplementary Table S4), the statement proved for any model that satisfies the stated hypotheses, the condition under which it applies to a numerical result, and the results of this study to which it applies. Levels of guarantee and identifiability per problem are given in Table 1; the formal statements are listed in Supplementary Table S7 and the scope of each guarantee in Supplementary Table S5.

Step of the method (modules)	Statement proved in Lean	Applies to a result when	In this study
Ranking by energy (1, 2, 11, 13)	The first-order output error when parameters are fixed is bounded by the discarded sensitivity energy; the bound transfers to transformed outputs	$R_{\mathrm{var}}(S) \geq \tau_R$ and the model is verified	Level 3: Alkan, Perelson
Selection with κ and VIF (7–9)	The greedy subset satisfies $\kappa \leq \tau_\kappa$ and $\mathrm{VIF} \leq \tau_{\mathrm{VIF}}$; discarding a rejected candidate permanently is correct; the linearised least-squares estimates are unique and stable	The computed κ and VIF are below the thresholds	All 31 subsets ($\kappa \leq 6.5$, $\mathrm{VIF} \leq 9.8$)
Admissibility threshold (3, 4, 20–22)	Theorem 1: rescaling an output change with cosine c leaves a relative error $\sqrt{1-c^2} = \tau_e$	Median $\cos\Delta \geq c$	Level 2 or 3: 17 subsets
Re-estimation (5, 12)	More re-estimated parameters never increase the fitting error; $e_{\mathrm{fit}} \leq e_{\mathrm{rel}}$; the first-order e_{fit} differs from the exact one by a second-order term	Always (exact e_{fit}); bound on the absolute error only (first order)	26 problems exact, 8 first order
Backward elimination (6)	Theorem 2: the result is admissible and no parameter can be removed	Always	All 31 subsets
Model hypotheses (14–18, 23–31)	The solution exists on the horizon, stays non-negative and is differentiable in the parameters	Checked for each model by the Lean kernel	28 models (27 of the 34 problems)
Identifiability (32)	If t_j^* is smaller than the distance to both bounds, θ_j is practically identifiable	I_j, γ and ε computed at the profile points satisfy the inequality	24 parameters; all of S in 7 problems

agreement to rounding error (Supplementary Note *Verified benchmark models*, Supplementary Table S8, with the complete data of one model in Supplementary Table S9). Supplementary Fig. S3 summarises how the two parts fit together: Python performs every computation on the benchmark problems, Lean proves the general statements of Supplementary Table S4 and checks the hypotheses of each model, and a conclusion of Supplementary Table S4 applies to a result of the paper only when both the model hypotheses are verified and the computed quantities satisfy the inequalities that the theorem requires.

Discussion

The central finding is that, for most published systems biology models with real data, the measured outputs can be reproduced by re-estimating only two biological parameters together with the measurement calibration parameters, and that, with the other parameters at their published values, the data also determine those two parameters. In the terminology of system identification this is a parameter subset selection[32] whose result is verified twice: the subset is sufficient to reproduce the outputs of the full model (admissibility) and its parameters are practically identifiable (bounded profile likelihood). With all parameters free, in contrast, most selected parameters are not identifiable, which is the sloppy-model picture[10]; our statement is more concrete and practical, because it refers to the outputs that were actually measured, under the actual experimental conditions and noise levels, and it is checked by re-estimation and by profiles rather than inferred from the sensitivity spectrum. It is a statement about output sufficiency and identifiability, not about predictive superiority: on synthetic data that the published estimate has not seen, the two-parameter models predicted neither better nor worse than the full models on average (we did not test equivalence), better in the problems in which the full model overfitted the training data and worse in others, and their advantage on real data reflects the information contained in the published values at which the other parameters are fixed.

The comparison with existing criteria clarifies what the method adds. When the remaining parameters are kept at their published values, subsets chosen by pivoted QR, Brun's collinearity index, Morris screening, profile likelihood or at random predict held-out experiments about equally well; many small subsets are adequate, and the test error alone cannot tell them apart. The criteria do differ in whether their subsets actually reproduce the full model: those of our method did so in every comparable problem, whereas the other criteria failed in 7–50% of the problems, and in 11–46% at a perturbation size that none of them, ours included, was tuned for. This reliability carried over to prediction on data that the published values do not inform, where our subsets reached the accuracy threshold significantly more often than those of pivoted QR and profile likelihood. Brun's collinearity index, which, like stage 1 of our method, combines sensitivity with a redundancy test, came closest and did not differ significantly in prediction; what distinguishes our method from it is that the subset is verified by re-estimation and reduced by backward elimination until no parameter can be removed, and that the verification makes failures visible (Beer, Crauste) instead of returning an untested subset. The ranking by sensitivity energy itself is not new: E_j is the diagonal of the Fisher information matrix for the scaled outputs and, up to a constant, the squared importance measure of Brun *et al.*[14]; the contribution lies in the verification that follows it. The two-stage design reflects a practical observation: ranking by energy (stage 1) was sufficient in only seven problems, whereas ranking by finite-perturbation effects (stage 2) was needed in most, because energy measures how strongly a parameter moves the outputs, not whether adjusting it can compensate for changes in other parameters.

The method is also practical. The number of sensitivity simulations grows linearly with the number of parameters and they can run in parallel; the complete method,

including the admissibility test, took from seconds to about half an hour per problem and ran on models with up to 294 parameters; profile likelihood, which repeatedly re-estimates the full model[5, 29], was as fast on small problems (our method was faster in 12 of 28) but took about an hour on 60 cores for Bachmann, the largest problem in which it was run, against 10 s for our method. It is applied before any new calibration, uses the published parameter values as its starting point, is insensitive to its numerical settings, and its subsets remain adequate for parameter changes of up to $\pm 50\%$ in most problems. Each step rests on statements proved in Lean 4. Other approaches answer different questions: likelihood-based observability analysis[7] assesses the uncertainty of predictions after calibration, Bayesian approaches[22] require many more simulations, structural identifiability analysis[2–4] addresses ideal noise-free data rather than a given experimental design, manifold-based sloppiness methods[28] identify emergent parameter combinations and rewrite the model rather than select individual parameters, balanced truncation[23, 24] reduces state variables rather than parameters, and hierarchical optimisation[31] eliminates calibration parameters during fitting without selecting among biological ones.

Several limitations should be noted. First, the analysis is local: sensitivities are computed at the published estimate θ_0, which is assumed to be a good fit to the data, and admissibility is tested around it; problems that are admissible with a small margin at $\pm 5\%$ (Bertozzi, Blasi, Elowitz, Lucarelli) are those that lose admissibility for larger changes. Second, because the non-selected parameters stay at θ_0, the reduced model's prediction of real held-out data benefits from the information that θ_0 already contains; a leakage-free nested protocol removes this benefit, and prediction on data that θ_0 has not seen is neither better nor worse than that of the full model on average, without a test of equivalence; the synthetic truths are generated around θ_0, so this test is also local. Third, the minimum subset size of two is a design choice; a single parameter already reaches the threshold in about half of the problems. Fourth, for the eight problems with the most expensive simulations the fitting error was computed to first order, and on small problems the admissibility test makes the method no faster than profile likelihood. Fifth, the method assumes smooth models, Gaussian noise with the recorded standard deviations and re-estimable calibration parameters; in five problems the recorded noise levels do not match the scatter of the data, in problems with many calibration parameters (Bachmann, Raimundez) the reduced problem is still sizeable, one benchmark problem with 9,169 experimental conditions was too expensive to analyse on its measured outputs with finite differences, and two problems had no admissible subset. Finally, the formal verification covers the mathematical steps of the method and the regularity of the translated models, not the numerical computations themselves, which we cross-checked against the reference simulations of the benchmark.

The social cognitive theory model, which motivated this work, illustrates the same behaviour outside molecular biology: with a single measured output the full estimation problem is ill-posed, and the conditioning test identifies two parameters that carry the information in that output. In practice, the method can be used as a preprocessing step: before calibrating a large model, select a small set of biological parameters sufficient to reproduce the measured outputs, verify that re-estimating

them reproduces the full model and that their profiles are bounded, keep the rest at literature values, and check the reduced model on held-out experiments. The threshold τ_e provides an explicit trade-off between accuracy and parsimony. Natural extensions include computing J with forward or adjoint sensitivity equations for genome-scale models, combining sensitivities at several operating points, and using the selected subsets to guide experimental design[25].

Methods

Models, data and simulation

We used the problems of the PEtab benchmark collection[8, 18] that provide real measurements, an SBML model and nominal parameter values (34 problems; Table 1). Each problem specifies the observables g_i, their transformation h (linear, log or $\log_{10}$), the noise standard deviation σ_i of each measurement, the experimental conditions (parameter or initial-value overrides) and, where applicable, pre-equilibration conditions. Models were simulated with libRoadRunner[26] (CVODE, relative and absolute tolerances 10^{-7} and 10^{-10}). Initial assignments that depend on parameters were re-evaluated for every parameter vector, condition-specific overrides were applied before simulation, pre-equilibration was run to steady state before the main simulation, and models with discontinuous inputs were integrated piecewise between discontinuities. Simulated observables were compared with the reference simulations distributed with the benchmark wherever available (Supplementary Note *Verification of the simulator*, Supplementary Table S10).

Social cognitive theory model

The model[16, 17] has six states x_i (constructs of social cognitive theory[15]) and obeys $\dot{x} = A(\theta)x + B(\theta)u$, $y = x_4$ (behaviour), with $A_{ii} = -1/\tau_i$, $A_{ij} = \beta_{ij}/\tau_i$ for the ten couplings $\beta_{14}, \beta_{21}, \beta_{25}, \beta_{31}, \beta_{34}, \beta_{42}, \beta_{43}, \beta_{45}, \beta_{46}, \beta_{54}$, and input gains $\gamma_{33}, \gamma_{35}, \gamma_{36}, \gamma_{57}, \gamma_{64}, \gamma_{68}$ divided by the time constant of the receiving construct (γ_{35} enters with negative sign; 22 parameters). The data are 31 daily records of one participant[17]: six measured inputs and daily steps as output, each standardised to zero mean and unit variance. The model is simulated from $x(0) = 0$ with inputs held constant over each day (exact zero-order-hold discretisation with the matrix exponential), and the output of day k is x_4 at the end of that day. θ_0 minimises the mean squared error by differential evolution, following the estimation framework of ref. [17] (strategy best1bin, population factor 4, 60 generations, Latin hypercube initialisation, no polishing; 5,368 function evaluations; seed 2026) within the bounds $\beta \in [-1.5, 1.5]$, $\gamma \in [0, 3]$, $\tau \in [0.3, 12]$, subject to stability ($\max \mathrm{Re}\, \lambda(A) \leq -10^{-3}$); the 100 runs with seeds 2026–2125 were used only to assess non-identifiability. The noise level σ is the standard deviation of the residuals at θ_0 (0.35 in standardised units). Because couplings may be negative, the reduced refit uses $\theta_j = \mathrm{sign}(\theta_{0,j}) \exp(v_j)$. All other settings are as for the benchmark problems. For validation, the reduced and full models were refitted to days 1–24 by least squares

from θ_0 within the same bounds and with the stability constraint, and the full model also by the same differential-evolution search; both were evaluated on days 25–31.

Measured outputs and parameter classes

For a parameter vector $\theta \in \mathbb{R}^p$ (in the units of the benchmark, excluding noise parameters), the vector of scaled outputs is $y(\theta) \in \mathbb{R}^m$ with entries

$$y_i(\theta) = \frac{h\big(g_i(x(t_i;\theta,c_i),\theta)\big)}{\sigma_i}, \qquad i = 1,\dots,m, \tag{1}$$

where $x(t;\theta,c)$ is the state trajectory under condition c and (t_i, c_i) are the time and condition of measurement i. Parameters that appear in the SBML model are biological (Θ_B); parameters that appear only in the observation functions and not in the model or the condition table (scalings and offsets) are calibration parameters (Θ_C). Only biological parameters are candidates for selection; calibration parameters are always re-estimated.

Sensitivities, energy and conditioning

The relative sensitivity matrix $J \in \mathbb{R}^{m\times p}$ is computed by central differences,

$$J_{:,j} = \frac{y(\theta_0 + \delta\theta_{0,j}e_j) - y(\theta_0 - \delta\theta_{0,j}e_j)}{2\delta}, \tag{2}$$

which approximates $\partial y/\partial \log\theta_j$. The energy of parameter j is $E_j = \|J_{:,j}\|^2 = (J^\top J)_{jj}$ and the fraction of energy captured by a subset $S \subseteq \Theta_B$ is $R_{\mathrm{var}}(S) = \sum_{j\in S} E_j / \sum_{j\in\Theta_B} E_j$. For a subset S, the columns of J_S are normalised to unit length, $Z = J_S \,\mathrm{diag}(\|J_{:,j}\|)^{-1}$; the condition number is $\kappa(S) = \sigma_{\max}(Z)/\sigma_{\min}(Z)$ and the largest variance inflation factor is $\mathrm{VIF}(S) = \max_j [(Z^\top Z)^{-1}]_{jj}$[13, 14]. Normalisation removes differences of scale between parameters, so that κ and VIF measure only the angles between sensitivity directions.

Selection

Greedy selection. Candidates are visited in a given order. A candidate j is accepted if $\kappa(S\cup\{j\}) \le \tau_\kappa$ and $\mathrm{VIF}(S\cup\{j\}) \le \tau_{\mathrm{VIF}}$; otherwise it is discarded permanently. Because κ and VIF can only increase when columns are added, a discarded candidate would also be rejected later. *Stage 1* visits biological parameters in decreasing order of E_j and stops when $R_{\mathrm{var}}(S) \ge \tau_R$. *Stage 2*, used if the stage-1 subset is not admissible, visits biological parameters in decreasing order of the finite-perturbation effect $s_j = \|y(\theta_0 + \eta\theta_{0,j}e_j) - y(\theta_0)\|$ and stops at the first admissible subset. *Backward elimination* visits the parameters of the selected subset and removes each one whose removal leaves the subset admissible.

Admissibility and re-estimation

Admissibility is evaluated on N random scenarios $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, with $u^{(k)}$ uniform in $[-1, 1]^p$. For each scenario let $\Delta y_{\text{full}} = y(\theta^{(k)}) - y(\theta_0)$. The relative error without re-estimation is $e_{\text{rel}} = \|\Delta y_{\text{full}} - \Delta y_S\| / \|\Delta y_{\text{full}}\|$, where Δy_S is the output change when only the parameters in $S \cup \Theta_C$ take their scenario values, and the cosine similarity is $\cos\Delta = \langle \Delta y_{\text{full}}, \Delta y_S \rangle / (\|\Delta y_{\text{full}}\| \|\Delta y_S\|)$. The relative error after re-estimation is

$$e_{\text{fit}} = \min_{\theta_S, \theta_C} \frac{\|y(\theta_S, \theta_C, \theta_{0,\overline{S}}) - y(\theta^{(k)})\|}{\|\Delta y_{\text{full}}\|}, \tag{3}$$

where $\theta_{0,\overline{S}}$ denotes the nominal values of all other biological parameters. The minimisation is carried out in $\log|\theta|$ within $[|\theta_0|/10,\ 10|\theta_0|]$ by a trust-region reflective least-squares method (SciPy[27], at most $100(|S| + |\Theta_C| + 1)$ function evaluations), from two starting points (the scenario value and θ_0); the error without re-estimation is also included, so that $e_{\text{fit}} \le e_{\text{rel}}$. Scenarios whose output change is below $\max(10^{-10}\|y(\theta_0)\|, 10^{-6})$, or in which the simulation fails, are discarded, and the number of valid scenarios is reported. Parameters with a negative nominal value keep their sign in the scenarios and in the re-estimation, which acts on $\log|\theta|$; a parameter with nominal value zero (four in Lang and two in Raimundez) is represented by 10^{-12}, so its relative perturbations and its column of J are negligible and, like any parameter with a zero or negligible column, it is in practice never selected (Supplementary Note *Random scenarios and large problems*). Supplementary Table S3 reconciles the parameter counts used throughout (estimated, biological, calibration and noise parameters, and the free parameters of each analysis). S is admissible if $|S| \ge k_{\min}$ and the median of e_{fit} over the scenarios is at most $\tau_e = \sqrt{1 - c^2}$. Because e_{rel} bounds the exact e_{fit} from above, the implementation also accepts a subset whose median e_{rel} is at most τ_e, which makes re-estimation unnecessary; with nonlinear re-estimation $e_{\text{fit}} \le e_{\text{rel}}$ in every scenario, so the rule $\min(\text{median}\, e_{\text{fit}}, \text{median}\, e_{\text{rel}}) \le \tau_e$ reduces to the median of e_{fit}, and it differs only for the first-order e_{fit} described below. For each admissible subset we also report a level of guarantee: level 1 when only this condition holds, level 2 when in addition the median $\cos\Delta$ is at least c, and level 3 when in addition $R_{\text{var}}(S) \ge \tau_R$; these levels are reported, not required. For the eight problems in which the cost of simulation made repeated re-estimation impractical (marked in Table 1), the first-order version $\min_c \|\Delta y_{\text{full}} - J_{S\cup\Theta_C} c\| / \|\Delta y_{\text{full}}\|$ was used. This equals the minimum relative error attainable if the model response were exactly linear in parameter changes; for a truly linear model it equals the exact e_{fit}, while for nonlinear models it is an approximation that replaces per-scenario optimisation with a single projection; its absolute error is bounded by a second-order term (Supplementary Table S7), but no bound on the relative error was evaluated for the individual scenarios. The admissibility decision for these eight models is therefore made by linear algebra alone and is not verified by nonlinear re-estimation. The columns of J for the calibration parameters are computed together with those of the biological parameters, because the first-order re-estimation uses $J_{S\cup\Theta_C}$.

Settings

Unless stated otherwise: $\delta = 0.01$; $\eta = 0.1$; $\tau_R = 0.89$; $\tau_\kappa = \tau_{\mathrm{VIF}} = 10$; $N = 15$ scenarios; $\varepsilon = 0.05$; random seed 42; $c = 0.9$, so $\tau_e = 0.436$; $k_{\min} = 2$. The sensitivity analysis changed one setting at a time: $\delta \in \{0.1\%, 5\%\}$; relative/absolute integrator tolerances $10^{-9}/10^{-12}$ and $10^{-6}/10^{-8}$; $\varepsilon \in \{0.01, 0.10\}$; $\tau_R \in \{0.80, 0.85, 0.95\}$; $\tau_\kappa = \tau_{\mathrm{VIF}} \in \{5, 30\}$; and $\tau_e \in \{0.10, 0.20, 0.30, 0.50\}$. It was run on the 26 problems that could be re-run completely fifteen times in reasonable time; the other eight (Alkan, Bachmann, Beer, Chen, Isensee, Lang, Lucarelli and Raimundez) are those with the most expensive simulations. To relate e_{fit} to the precision of the data, the remaining error of each scenario was also expressed per measurement in units of the recorded noise, $e_{\mathrm{fit}}\|\Delta y_{\mathrm{full}}\|/\sqrt{m}$ (the outputs are already divided by σ_i), and its median over the scenarios was reported. Single-parameter subsets were tested on 27 admissible problems (all except the four with the most expensive simulations, Bachmann, Chen, Isensee and Raimundez): the ten biological parameters with the largest E_j and the ten with the largest s_j were each tested alone, with Θ_C re-estimated, on the same 15 scenarios at $\pm 5\%$ and without the requirement $|S| \geq 2$; the admissible candidate with the smallest median e_{fit} was then validated like the selected subset.

External validation

Measurements were split into training and test sets by experimental condition (every third condition after sorting, or the second of two conditions, used for testing) or, for problems with a single condition, by time (the last third of the distinct measurement times). Parameters that only affect test conditions were excluded from re-estimation. The reduced model (re-estimating $S \cup \Theta_C$) and the full model (re-estimating all parameters) were fitted to the training data from θ_0 with the same least-squares method and bounds, and evaluated by the mean squared noise-scaled residual $\chi^2/n = \frac{1}{n}\sum_{i \in \mathrm{test}}(y_i(\hat{\theta}) - \tilde{y}_i)^2$, where $\tilde{y}_i$ are the scaled measurements. In the synthetic experiment, the truth θ^* had each estimable parameter drawn uniformly within $\pm 20\%$ of θ_0 (parameters acting only in test conditions stayed at θ_0), data were simulated with Gaussian noise of standard deviation σ_i, and $e_{\mathrm{pred}} = \|y_{\mathrm{test}}(\hat{\theta}) - y_{\mathrm{test}}(\theta^*)\|/\|y_{\mathrm{test}}(\theta^*) - y_{\mathrm{test}}(\theta_0)\|$; the effect of the parameter change was considered larger than the noise when χ^2/n of θ_0 on the synthetic test data was at least 2. The experiment was repeated with 20 seeds (42–61), each defining a different truth and noise realisation; the full model was not re-estimated for the two problems with more than 100 free parameters after removing those that act only in the test conditions (Chen and Raimundez; Bachmann has 101 parameters that affect the outputs but 84 free ones, and its full model was re-estimated). The experiment was repeated with truths within $\pm 5\%$ and $\pm 50\%$ of θ_0 (seeds 42–51, ten seeds), and compared with the same ten seeds at $\pm 20\%$. The true signal of a case is the root mean square, over the test measurements, of $y_{\mathrm{test}}(\theta^*) - y_{\mathrm{test}}(\theta_0)$, which is in units of σ because the outputs are scaled by the noise; it is computed from the truth, without noise, whereas the criterion "larger than the noise" above uses the noisy synthetic data. Problems whose test χ^2/n at θ_0 was below 0.01 (Giordano, Perelson,

Rahman) reproduce the data to far below the recorded noise level and are not shown in Fig. 3a. Test errors were compared through the ratio of the reduced to the reference χ^2/n; ratios between 0.95 and 1.05 were considered equivalent. For the leakage-free nested validation (Supplementary Note *Leakage-free nested validation*), the full model was first re-estimated on the training data (yielding $\hat{\theta}_{\mathrm{train}}$), and the reduced model was then re-estimated with the non-selected parameters fixed at $\hat{\theta}_{\mathrm{train}}$ and search bounds derived from $\hat{\theta}_{\mathrm{train}}$, for 29 problems (Bertozzi and Blasi excluded; for Chen the columns of the finite-difference Jacobian were evaluated in parallel and the integrator tolerances were $10^{-6}/10^{-8}$).

Comparison with other selection criteria

Each criterion selected the same number $k = |S|$ of biological parameters as our method, from the training data, and the calibration parameters of the problem were always added. *Pivoted QR*: column-pivoted QR decomposition of the sensitivity matrix of the training outputs at θ_0 with columns normalised to unit length[32]; the first k pivots. *Brun*: dimensionless sensitivities $\theta_j\,\partial y_i/\partial\theta_j$ on the training outputs, importance δ_j^{msqr}, and among the 15 most important parameters the k-subset with collinearity index $\gamma \le 10$ and the largest determinant measure ρ (the smallest γ if none satisfies the bound)[14]. *Morris*: ten trajectories (seed 42) on a four-level grid of $\log|\theta_j|$ within $[\theta_0/10,\, 10\theta_0]$, elementary effects measured by the norm of the change of the training outputs, and the k parameters with the largest μ^*[12]. *Profile likelihood*[5, 29]: the full model was first re-estimated on the training data; each biological parameter was then fixed at five values on each side of the optimum (up to a factor of 10, within the same bounds as all other fits) and all other parameters were re-estimated; whenever a point of a profile improved the optimum by more than one unit of χ^2, the full model was re-estimated from that point and the profiles were repeated (at most three times; ten for Raia). A parameter was classified as identifiable when its profile exceeded $\Delta\chi^2 = 3.84$ (95%, one degree of freedom) on both sides, and the k parameters with the narrowest confidence intervals were selected. It was run for the 26 admissible problems with at most 60 free parameters and, in a separate run with the profile points distributed over 60 cores, for Bachmann and Lucarelli (90 and 72 free parameters); it was not run for Chen and Raimundez (155 and 126 free parameters), whose simulations are the most expensive. The criteria were implemented as they are usually applied and therefore differ in some numerical settings (data used to select, derivative scheme and step, candidate set, normalisation and computational budget); these differences are listed in Supplementary Note *Brun and Morris selections*. The *calibration-only* baseline re-estimated Θ_C alone, and the *random* baseline used the median test χ^2/n over 20 random k-subsets. Every subset was then evaluated with the external validation described above, with the admissibility test of our method (the same 15 scenarios and the same rules, at $\pm5\%$ and at $\pm20\%$) and with the synthetic experiment (20 seeds). The problems entering each comparison follow from these requirements. Admissibility was compared in 28 of the 31 admissible problems: Bertozzi was excluded (see Statistics), Blasi has no training/test split on which the other criteria could select, and Chen was not included because the sensitivity computations of Brun and Morris on its training

outputs would take hours. Profile likelihood was compared in 27 of them (all except Raimundez). The synthetic comparison used the 24 problems with at least one seed in which all five criteria had a result and the parameter change exceeded the noise: of the 28, Raimundez lacks the profile-likelihood subset, and Giordano, Laske and Lucarelli have no seed with signal. The computation time of the complete method was measured by running it from scratch, without cached results, with the sensitivity computations distributed over 16–64 cores; all timings were obtained on a 128-core server shared with other computations and are therefore upper bounds.

Statistics

Comparisons between criteria were made over problems. For real data and for the synthetic prediction error, the statistic of each problem was the logarithm of the ratio of the errors of two criteria (for synthetic data, its median over seeds) and the two-sided Wilcoxon signed-rank test was applied. For the fraction of synthetic cases reaching $e_{\mathrm{pred}} \leq \tau_e$, 95% confidence intervals of the difference between two criteria were obtained by resampling whole problems, with all their seeds, 10,000 times (seed 0), so that seeds of the same problem are not treated as independent. The two-sided p value of the bootstrap is twice the smaller of the fractions of resampled differences at or below and at or above zero; in the Wilcoxon test, problems with equal errors were discarded (zero method of Wilcoxon). Admissibility, one outcome per problem, was compared with the exact McNemar test on the problems in which only one of the two subsets was admissible. The comparison of admissibility and of the synthetic success rate with the four existing criteria is the primary analysis; within each family of four comparisons (admissibility at one perturbation level, or synthetic success rate) p values were adjusted with the Holm procedure, and the comparisons on real data are exploratory. The absence of a significant difference is not evidence of equivalence; no equivalence margin was tested. Bertozzi was excluded from the comparison of admissibility because our method selected its parameters on all measured outputs, which in this problem include the test conditions, whereas the other criteria selected on the training outputs.

Identifiability of the selected parameters

For each problem with an admissible subset, the reduced problem ($S \cup \Theta_C$ free, other parameters at θ_0) was fitted to all measurements from θ_0, with the method and bounds of the external validation. For each selected parameter θ_j, $\log|\theta_j|$ was fixed at displacements of 0.05 to 4.6 from the optimum and the other free parameters were re-estimated; points outside the bounds $[|\theta_0|/10, 10|\theta_0|]$ of θ_j were not used, and if a point improved the optimum the reduced problem was re-fitted from it and the profile repeated (at most three rounds). θ_j was classified as identifiable if $\Delta\chi^2$ exceeded 3.84 on both sides, practically non-identifiable if on one side, and flat otherwise[5]; a robust variant scaled the threshold by $\max(1, \chi^2_{\mathrm{min}}/\mathrm{d.f.})$. Let $r(\delta)$ be the vector of noise-scaled residuals of all measurements when the logarithms of the free parameters move by δ from the optimum, $r_0 = r(0)$, so that $\chi^2 = \|r\|^2$, and let A be its sensitivity matrix at the optimum (central differences, step 0.01 in $\log|\theta|$).

The profiled information I_j was computed as the squared norm of the component of the column of A for θ_j orthogonal to the other columns, which equals $1/[(A^\top A)^{-1}]_{jj}$ when $A^\top A$ is invertible; it is the output change, in noise units, that a unit change of $\log|\theta_j|$ produces when the other free parameters compensate as well as possible. The residual gradient γ is the norm of the projection of r_0 onto the range of A (zero at an exact optimum), and ε is the largest nonlinear remainder $\|r(\delta) - r_0 - A\delta\|$ over the computed profile points. The Lean theorem `practically_identifiable_grad` states that if $|\langle r_0, A\delta\rangle| \le \gamma\|A\delta\|$, the nonlinear remainder is at most ε, and $t^* = (\varepsilon + \gamma + \sqrt{\gamma^2 + 3.84 + 2\|r_0\|\varepsilon + 2\gamma\varepsilon})/\sqrt{I_j}$ is smaller than the distance to the bounds on both sides, then θ_j is practically identifiable; the remainder can be evaluated only at points, so this condition was checked at the computed profile points. The complete mathematical derivation, the profiles of every problem and the values of all quantities for every parameter are given in Supplementary Note *Identifiability of the selected parameters*. The profiles were computed for all 31 problems with an admissible subset: those of Bachmann, Chen, Isensee and Smith on a 256-core server, that of Raimundez on a 128-core server, and the other 26 on an 8-core workstation (libRoadRunner 2.10.0 on the servers and 2.9.2 on the workstation).

Robustness patterns

For each admissible problem, the final subset was kept fixed and e_{fit} was evaluated at perturbation sizes $\varepsilon \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$ with 15 scenarios per level. The fifteen directions $u^{(k)} \in [-1, 1]^p$, with independent uniform components, are drawn once per problem with a fixed seed (42) and reused at every level, $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$; the directions are stored with the results. To check that the patterns do not depend on these particular directions, the evaluation was repeated with new directions at each size (seed 42 at $\pm 5\%$, which reproduces the selection, and $42 + 1000i$ at the i-th other level). Patterns were assigned from the median e_{fit} at each level as described in Results; the pass rate is the fraction of valid scenarios with $e_{\text{fit}} \le \tau_e$. As in the selection, e_{fit} was computed to first order for the problems marked in Table 1. For Chen, between 8 and 13 of the 15 scenarios per level could be simulated in the run shown in Fig. 5 (7 to 13 over all our runs, because some simulations fail in a run-dependent way); the remaining scenarios failed to integrate and were discarded.

Formal verification

The formal development uses Lean 4 (version 4.23.0) and Mathlib[19, 20] and has two parts (Table 3 and Supplementary Fig. S3). The first proves, for any model that satisfies stated hypotheses, the mathematical results on which each step of the method relies (Supplementary Table S4, steps 1–22). The second establishes those hypotheses for concrete benchmark models. For this, each PEtab problem was translated automatically into a Lean file (`lean_translation/sbml_to_lean.py` of the reproducibility package): the rate laws and stoichiometry of the SBML model and the nominal parameter values and initial conditions of the PEtab tables are written as Lean expressions in which every number is an exact rational. Verified checkers, Boolean functions proved sound in Lean, then test that every rate law is defined and

smooth on a domain containing the trajectory, that concentrations cannot become negative (quasi-positivity) and that the growth of the vector field is bounded so that the solution exists on the simulated horizon (linear growth, or a Riccati bound on a finite horizon for Crauste). The checkers are evaluated by the Lean kernel itself (`decide +kernel`), so their results are part of the proof, and each model ends with a theorem stating that it is smooth, that its solution exists and remains non-negative on the horizon, and that the solution is differentiable with respect to the parameters, which are the hypotheses of steps 14–22. This was achieved for 28 benchmark models (27 of the 34 problems analysed, and Froehlich); the reasons why the other seven problems were not verified are given in Supplementary Note *Verified benchmark models*. Because the theorems are only as relevant as the translation is faithful, the translated vector fields, parameter values and initial values were compared independently with the SBML models evaluated by libRoadRunner at the nominal state and at five random states; they agree to rounding error (largest relative difference 10^{-12}), and changing a single number of a Lean file by 1% is detected. All files compile without `sorry`, and `#print axioms` reports only the standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`). The numerical analyses of this paper (J, R_{var}, κ, VIF, $\cos\Delta$, e_{fit} and the validation) were carried out in Python in double precision; a formal conclusion applies to a numerical result when the model's hypotheses are verified in Lean and the computed quantities satisfy the inequalities that the corresponding theorem requires (Supplementary Table S5).

Use of large language models

An AI assistant (Claude, Anthropic) was used to help write and debug analysis code and Lean proofs and to edit the manuscript text. All code, proofs and results were checked by the authors, who take full responsibility for the content.

Data availability. The PEtab benchmark problems analysed in this study are publicly available in the Benchmark-Models-PEtab collection (https://github.com/Benchmarking-Initiative/Benchmark-Models-PEtab, commit `fcbddf1`). All results generated in this study (selected subsets, external validation, synthetic prediction experiments, comparison with other selection criteria including the profile likelihoods, identifiability of the selected parameters, robustness, single-parameter subsets, sensitivity to the method settings, simulator check, computation times and the checks of the Lean models), one file per benchmark problem with a data dictionary, are included in the reproducibility package archived on Zenodo (doi:10.5281/zenodo.23338175) and in the folder `reproducibility/` of https://github.com/monicammr/archivos. The participant's step-count data of the social cognitive theory model are available from the corresponding author on reasonable request.

Code availability. The reproducibility package (Zenodo, doi:10.5281/zenodo.23338175; folder `reproducibility/` of https://github.com/monicammr/archivos) contains the Python code of the method and of every analysis, the programs that produce every figure and table of the paper from the stored results, the translator of the SBML models into Lean, and all Lean 4

proofs (folder `lean_proofs/`, a self-contained project that compiles with a single command, `./verify.sh`, and checks that no proof uses `sorry` or additional axioms). Its manual (`README.md`) describes three levels of checking: regenerating the figures and tables from the stored results, compiling the proofs, and re-running the analyses from the PEtab problems (the analyses of the social cognitive theory model additionally require the participant's data, available on request); Supplementary Note *Reproducibility package* summarises them.

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Ethics declarations. Ethics approval and consent to participate: The step-count data used for the social cognitive theory model were collected in a previously published study[17] approved by the Institutional Review Board of that study. The participant provided written informed consent for the use of their anonymised data in subsequent analyses.
Consent for publication: The participant whose data underlie the social cognitive theory analysis gave consent for publication of anonymised results.

Additional information. Competing interests: The authors declare no competing interests.
Correspondence and requests for materials should be addressed to M.M.R.
Supplementary information accompanies this paper (Supplementary Notes, Tables and Figures); Supplementary Note *Concluding remarks* summarises its contents.

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